MCU unit

U2
ESP32_FEATHER_V2

VBAT 1 VBAT
2 EN
VUSB_5V TP4 3 VBUS
A0_row 4 IO13_A12
A1_row 5 IO12_A11
A2_row 6 IO27_A10
1A0 7 IO33_A9
1A1 8 IO15_A8
2A0 9 IO32_A7
2A1 10 IO14_A6
USB_PD_SCL 11 SCL
R15 USB_PD_SDA 12 SDA
10k 0603
R13 10k 0603
3V3
R14 10k 0603
GND

RESET 28
PWR_FLAG
3V3 27 3V3 TP6
NC 26
GND 25 GND TP2
A0_DAC2 24 A0_col
A1_DAC1 23 A1_col
A2_I34 22
A3_I39 21
A4_I36 20
A5_I04 19 2E
SCK 18 A2_col
MOSI 17 2A_2B
MISO 16 1E
RX/IO7 15 board_on
TX/IO8 14 24V_EN
I37 13

3V3
R8 100k 0603
R2 10k 0603
BUTTON_IN
C7 22n 50V 0603
GND

24V Boost

24V_REGIN
TP8
IC1
TPS55340PWP
PGND_3 14
PGND_2 13
PGND_1 12
NC 11
FREQ 10
FB 9
COMP 8
SW_1 1
SW_2 2
VIN 3
EN 4
SS 5
SYNC 6
AGND 7
EP 15
GND
SPM6550T-3R3M-HZ
1 24V_EN L2 2
D1 2A 40V
TP9
R7 12 1206
PWR_FLAG
24V_BOARD
TP10
R3 187k 0603
FB_B
R4 10k 0603
C8 UWT1V470MCL1GS
R12 0 0603
VUSB
24V_BOARD
In case VUSB=20V constant
R5 60.4k 0402
R6 3.3k 0603
C1 220n 16V 0603
C3 1.2n 50V 0603
C2 47n 50V 0402
24V_REGIN
C4 100n 16V 0603
C5 UWT1V470MCL1GS

MH2
Hole_2.7mm_M2.5_DIN965

MH1
Hole_2.7mm_M2.5_DIN965

Connectors

J4
2xSolderpads
PWR_FLAG
VUSB TP7
TP5 GND
adafruit USB C PD 5807

J8 2xSolderpads
USB_PD_SDA
USB_PD_SCL
J3 2xSolderpads
BUTTON_IN
GND

Only BOM
J6 SSA-116-S-G
J5 SSA-112-S-G

R9 0 0603
PWR_FLAG
VBAT
J2 S2B-PH-K-S__LF__SN_
PWR_FLAG
GND

J1
GND GND
Vboard 3
A0_row 5
A1_row 7
1A1 9
A0_col 13
A2_col 15
2A1 17
TP1 2E 19
A2_row 6
1A0 8
1E 10
A1_col 14
2A0 16
2A_2B 18
24V_BOARD 26
(1E not really used when updating screen. Could be set high at init)
Vboard
R11 10k 0603

VSYS
R1 10k 0603
board_on 1
Q2 DMG2305UX-7
Vboard

DMG2305UX-7 Q1
VBAT 3
2 VSYS TP3
1 VUSB_5V
System voltage selector. If VUSB_5V is present, then Vsys = VUSB_5V − 0.5V. Else, Vsys = Vbat
D3 2A 40V
VUSB_5V
PWR_FLAG
VSYS
Could maybe switch to diode with lower leakage current

DMG2305UX-7 Q3
VSYS 3
2 24V_REGIN
R10 187k 0603
VUSB
GND
24V reg input voltage selector. If VUSB is present (5-20V) then V24V_REGIN = VUSB-0.5V Else, V24V_REGIN = VSYS
D2
VUSB
PWR_FLAG
24V_REGIN
2A 40V
Pull-down in order for VUSB not to float in case unconnected

3.8−40V in to 5V out
(Supply for esp32 feather and board logic only)

C6 100n 16V 0603
IC2
AP64351SP-13
VUSB
C9 10u 10V 0805
C10 10n 100V 0603
BST 1
VIN 2
EN 3
SS 4
SW 8
GND 7
COMP 6
FB 5
E_PAD 9
L3
NRS4018T6R8MDGJ
1 2
C15
R18 13.7k 0603
C13 3.3n 50V 0402
C14 33p 50V 0402
R17 115k 0603
R16 22.1k 0402
PWR_FLAG
VUSB_5V
C11 22u 10V 1206
C12 22u 10V 1206
3.8−40V input, 5V output Buck. Should work fine with 5V in, 5V out. (100% duty)

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